HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F78AA

ACTUARIAL AND FINANCIAL MATHEMATICS A

Semester 1 2018/19

Duration: Two Hours

Total marks: 60

ANSWER ALL QUESTIONS

Electronic calculators approved by the University may be used.

1. The effective rate of interest is 8% per annum. Cashflows are as follows (the time unit is one year):

Time	Cashflows
0	+1000
1.5	-500
2	+100
2.5	-500
4	-500

Cashflows only take place at the given times.

For these cashflows, calculate:

(a) The total accumulation value at time $t = 5$ years. [3 Marks]

(b) The total present value at time $t = 0$ years. [1 Mark]

(c) The total value at time $t = 2$ years. [1 Mark]

[Total 5 Marks]

2. ABC investment fund has provided you with the following information about its fund performance over the last three years:

Calendar Year	Fund Value on 1 January	New Cash Invested on 31 December
2015	14,000,000	1,500,000
2016	16,900,000	500,000
2017	16,893,000	2,000,000
2018	20,920,160	–

Calculate the Time-Weighted Rate of Return over the period 1 January 2015 to 1 January 2018.

[Total 4 Marks]

3. (a) Writing down all steps, show from first principles that

$$(I\ddot{a})_{\overline{n}|} = \frac{\ddot{a}_{\overline{n}|} - nv^n}{d}.$$

[4 marks]

(b) Calculate the present value, at a rate of interest of 3% per annum effective, of an annuity where £10 is paid at the beginning of the first year, £12 is paid at the beginning of the second year, £14 is paid at the beginning of the third year and so on, with payments increasing by £2 per annum until the payment stream ends after 20 payments.

[6 marks]

[Total 10 Marks]

4. On 1 January 2015, a corporation issued a number of bonds with a coupon of 4% per annum payable annually in arrears on 1 January until the bond's redemption on 1 January 2043. Each bond has a nominal of £100 and will be redeemed at par.

The price of the bond at the time of issue resulted in a gross redemption yield of 4% per annum.

An investor, John, bought the bond at the issue date. John pays income tax at the rate of 20% and capital gains tax (CGT) at the rate of 30%.

CGT is zero if there is a *loss* of capital.

On 1 January 2018, just after payment of the coupon due on that date, John sold the bond to another investor, Kelly. Kelly pays income tax at the rate of 25% per annum, and CGT at the rate of 40%.

If Kelly holds the bond until the redemption date, then she will earn a yield, after taxes, of 3.2% per annum.

(a) Calculate the price of one bond on the issue date, 1 January 2015.

[1 Mark]

(b) Calculate the price per bond paid by Kelly on 1 January 2018.

[6 Marks]

(c) Show that the yield after tax realised by John, *on completion of the transaction with Kelly*, lies between 1.8% and 1.9% per annum.

[5 Marks]

[Total 12 Marks]

5. You have borrowed £2,000 from a bank to invest in a project.

The bank charges a rate of interest of 8% to borrow money, and pays 6% interest on money that is deposited in the bank (a bank deposit account).

The project produces the following cashflows:

- £200 at the end of the first year;
- £210 at the end of the second year and increasing at the rate of 5% per annum compound thereafter;
- the final payment is at the end of the 20th year.

(a) You will use the cashflows from the project to repay the bank loan as soon as possible.

i. Show that the amount of the 20th repayment to the bank is £505.39.
[1 Mark]

ii. Calculate the Discounted Payback Period for this project.
[4 Marks]

iii. Calculate the accumulated profit at the end of 20 years.
[5 Marks]

(b) Suppose instead that you pay each of the project cashflows into a bank deposit account until the end of 20 years. You then pay off the initial loan plus interest.

i. Calculate the accumulated profit at the end of 20 years.
[3 Marks]

ii. Explain why your profit in (b)(i) is less than the profit in (a)(iii)?
[1 Mark]

[Total 14 Marks]

6. Suppose you have two series of cashflows:

- cashflows of $a_1, a_2, \ldots, a_n$ are payable at times $t_1, t_2, \ldots, t_n$ respectively, and
- cashflows of $b_1, b_2, \ldots, b_m$ are payable at times $s_1, s_2, \ldots, s_m$ respectively,

for strictly positive integers n and m. Suppose that the yields on both sets of cashflows exist and are equal to the same value $i^\star$.

(a) Show that if you combine the two sets of cashflows into a single transaction, then $i^\star$ is also a solution of the equation of value of the new, combined cashflows.

[6 marks]

(b) Setting $n = m = 2$, find an example to show that $i^\star$ need not be the unique solution of the equation of value that is greater than -1 for the combined cashflows.

[Hint: consider the cashflows $a_1 = +1$, $a_2 = -1.1$ at times $t_1 = 0$, $t_2 = 1$, respectively, and $b_1 = -1.1$ and $b_2 = +1.21$ at times $s_1 = 2$, $s_2 = 3$, respectively. Write down the equations of value of each cashflow and of the combined cashflow. Show that $i^\star$ is a solution of all three equations but, also, that the equation for the combined cashflows has another solution.]

[6 marks]

(c) State Steffensen's Theorem and give an economically important situation where it applies.

[3 marks]

[Total 15 Marks]

END OF PAPER